Sahaj Bhadja

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Summary

Software engineer with experience in **C/C++** and **systems programming**. Strong foundation in **distributed systems**, **PostgreSQL**, and database internals. Proficient in data structures, algorithms, concurrency, and threading. Excited about building production-grade **distributed databases**.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Distributed Systems, Database Systems, Operating Systems, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Data Structures, Algorithms, Operating Systems, Database Management

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Developed high-performance **C++ components** for data processing systems, writing comprehensive **unit tests and integration tests** to ensure correctness and stability.
- Optimized **PostgreSQL query execution** paths and improved database throughput, troubleshooting performance issues in complex **distributed** query planning.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Built **C++ systems** with multi-threading and **concurrency** control, participating actively in code reviews and maintaining high code quality standards.
- Collaborated with cross-functional engineering teams to solve complex problems, improving **scalability** as data volumes grew.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Developed **C++ modules** for distributed data pipelines, implementing efficient **data structures and algorithms** for storage and retrieval operations.
- Added **regression tests** to ensure stability, demonstrating strong problem-solving skills in collaborative **distributed team** environment.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Built **C++ systems** for real-time data **replication** and synchronization, implementing **transactional** subsystems with fault tolerance guarantees.
- Worked with **PostgreSQL** internals including query layer, storage, and execution paths, contributing to **open source** database tooling.

DataCore Systems **India**
Systems Engineer Intern *January 2023 – April 2023*

- Developed **C/C++ code** for database storage engine components, implementing **threading** and concurrency primitives for distributed query execution.
- Contributed to **open source** database projects, writing automated tests and improving **distributed SQL** capabilities across releases.

Skills

Languages: C, C++, Java, Python, SQL
Databases: PostgreSQL, distributed databases, SQL query optimization, storage systems
Systems: Distributed systems, concurrency, threading, replication, transactions
Fundamentals: Data structures, algorithms, operating systems, fault tolerance, Git

Projects

MiniDB Code | *C++, PostgreSQL, Distributed Systems*

- Built a **distributed SQL database** engine in C++ with PostgreSQL-compatible query layer, implementing **replication, transactions**, and fault tolerance mechanisms.

QueryOptimizer | *C++, PostgreSQL, Algorithms*

- Developed **query execution** optimizer improving throughput by 40%, implementing efficient **data structures** for distributed query planning and execution paths.

Chat2SQL Code | *PostgreSQL, SQL, Python*

- Created **SQL query** generation tool demonstrating deep understanding of **PostgreSQL** and database fundamentals with comprehensive test coverage.